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An Overview of Requirements Elicitation Techniques in Software Engineering with a focus on Agile Development

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Abstract- In software requirement engineering, main focus is on eliciting information and extracting the main requirements of user. For extracting the requirement of the user, selection of appropriate tool/technique for specific situation is mandatory. Requirement elicitation is a difficult phase of requirement engineering. Elicitation includes traditional, collaboration, cognitive, innovative and contextual techniques. According to the nature of project and situation, one elicitation technique is not sufficient. This paper inspects that how requirement elicitation and its different tools/techniques work and what are the drawbacks of tools/techniques on the basis of particular situation that are demanded by the client. Agile model is the most common model of software development. Therefore, in this study tools/techniques of elicitation that are used in agile development phase have also been discussed in detail. This paper helps the analyst and project managers to select the appropriate tool and technique for a particular situation.

Keywords--Requirement Engineering (RE), Elicitation, Agile development, Software, requirements, Stakeholders

1. INTRODUCTION

Requirement Engineering (RE) is a process of extracting all information about the system like documentation, working and needs and desire of the stakeholders [4], inadequate requirements become the reason for software or project failure [5]. There are seven phases of Requirement Engineering (RE) which are: Inception, Elicitation, Elaboration, Negotiation, Specification, Validation, and Requirement Management. In order to extract accurate information about the system elicitation phase is used [6].

Elicitation is a basic or starting phase of RE [7]. Requirement elicitation is the process of eliciting information rather than collecting and this is a very critical thing like this is the point where the project is made according to the gathered information [8]. In Elicitation, requirements are gathered using different tools/techniques [9]. Elicitation is a critical and difficult process for gathering the requirements [10] [11], so it also affects the quality of the software, and sometimes it leads to failure [12] [13]. The essential part of any project is gathering information because this is the base of any project if this won't be effective then the whole project will be useless, a viewing process called a human-centered journey means gathering participants' experiences [14]. There are several types of tools/techniques are proposed for requirements elicitation: traditional techniques, collaborative techniques, contextual techniques, cognitive techniques, and innovative techniques.

The most common and old techniques for the elicitation process are called traditional techniques which include: Questionnaires, Interviews, and Meetings. [15]. Techniques that involve a group of stakeholders who apply their individual ideas for a set of decisions and they belong to different fields are called collaborative techniques which include brainstorming, Prototyping, Joint Application Development (JAD). Contextual techniques are those techniques in which requirements are collected in the working environment like ethnography, observation [1]. Cognitive techniques include card sorting, reporter grids, laddering. Which are based on multidisciplinary for gathering information at human thinking level, the fields include business management, psychology, and cognitive science. [16]. Despite all the tools/techniques which are discussed above about Elicitation in Requirement Engineering (RE) are not enough to solve several issues that's why the innovative technique is used which is through away paper prototype which is the visual representation of the system, this paperwork can't be used for a long time.

What is the preliminary and modern elicitation tool/technique for particular requirement gathering situations? And how to select the best tool/technique for the elicitation phase? To accomplish the above-mentioned goals of the research, the following research questions are synthesized.

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RQ1: What are the limitations of requirements elicitation techniques and why they are not applicable/inefficient?
RQ2: What are the suitable tools/techniques for agile development in the elicitation phase?
In this paper different tools/techniques have been reviewed with their working, limitation, type. Section 1 contains the research methods, in which each of the aforementioned questions are discussed against their article, search strategies to perform each identified process, and also highlights the inclusion & exclusion criteria regarding the search papers. Section 2 answers the research questions. Section 3 discuss briefly the purpose of the search; also discuss some techniques or tools that are used in literature. In Section 4 we concluded our findings.

II. Literature Review

In the research process, the 1st stage of Searching Criteria lies in Google scholar, IEEE Explorer, and Research Gate. In these criteria, we've found 100 Articles from which 80 papers were related to elicitation. In the 2nd stage of Searching, we've found 80 research papers in the portion of references, in which 38 articles are selected. By gaining the Search Material from these 38 articles, we've finalized our paper in Fig. 1.

The elicitation phase includes several types of tools/techniques like cognitive, collaborative, traditional, contextual, and innovative are discussed in this paper. Basically, these tools/techniques help us to tackle problems of various situations. For instance, if we have to design a visual representation of the system, we use innovative techniques. As well as innovative techniques to handle small and medium-sized projects. In this technique, users are told to draw the features with a pen/pencil. While when we have to gather or elicit information about application domain and generic knowledge in a group then collaborative tools/techniques are used. The technique which is most commonly used to extract data or information by asking questions or reading different scenarios is the traditional technique. For a collection of complete information about stakeholders like their working, environment, patterns contextual techniques are used. The techniques that are used to collect the data/information about projects from multiple fields like education, business management. All tools/techniques have been briefly discussed in Table 2. This paper also includes those tools/techniques in agile development for eliciting or gathering information. Tools/techniques include user story, Joint Application Development (JAD), ARAM, and DiCOT. They all are discussed in detail with their work and limitation in Table 3.

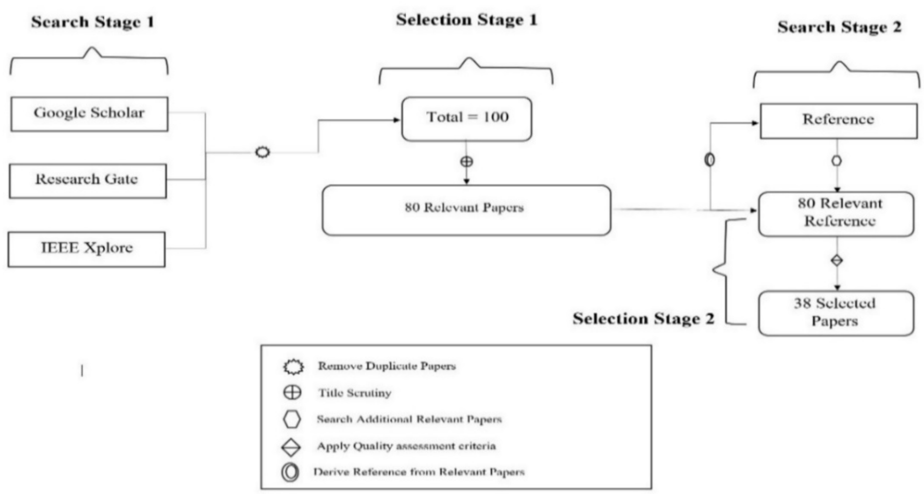

Figure 1: Search Process

The guidelines of Kitchen ham et al [17] were used in this literature review. First of all, in search stage 1 article were found by doing simple searching regarding elicitation techniques. In this stage, 100 articles were found but 80 articles were related to our topic furthermore in stage 2 these useful 80 papers were checked according to exclusion and inclusion criteria that have been mentioned below in table 1. After reviewing 80 articles, 38 articles were selected as more informative and useful for this research paper.

Inclusion Criteria:

- All papers in this research article are in the English language.
- All the relevant Papers lies in the criteria of different models, their tools and techniques, and their limitation.

- All the research papers are journals or research papers and duration includes 2010-2021.

Exclusion Criteria:

- Research papers except the English language.
- All research papers before last 11 years.
- Those articles that behave as survey
- All papers from the web.

Search Engine: The research articles were taken from the following scientific digital libraries:
Google Scholar, IEEE explorer, Research Gate.

***Search Strings**:* The search strings or queries which were used to extract the required articles. (“Elicitation” OR “Requirement Collection” OR Gathering AND (“Tools” OR “Techniques” OR “Methodology” OR “Strategies”)

A. Elicitation Techniques (RQ1): First, we are elicited tool/techniques of elicitation phase of requirement engineering. Discussion of these tools and techniques are presented in Table: 1

Table 1: Techniques and tools for Requirement Elicitation

Tool/ Technique	Working	Limitation	Applicability	Type	Ref.
Interview	It is basically a communication that takes place between two or more stakeholders. It can be open and closed-ended. It is best approach.	It is applicable to a small group of people. Inconsistency may affect the information.	Small group of stakeholders. Medium size projects.	Traditional	[7] [18] [19] [20] [21] [22] [23] [24]
Questionnaire/ Survey	It is used for gathering information by asking questions. This technique is used for the collection of large data from a number of stakeholders. It is the cheapest way in less time at a low cost.	It doesn’t need a brief explanation of any question; it is only concerned with particular information. No flexibility in stakeholder’s language or idea.	Large group of stakeholders. Large size projects.	Traditional	[25] [26] [21] [27] [28] [29]
Introspection	It is also an easy and cheap technique. Analysts build requirements according to their experience that is what requirement is useful to stakeholders or other users.	Experience of analyst can’t reach the level of user’s main requirement. It is applicable only at the start of the project.	Large group of stakeholders. Large size projects.	Traditional	[12] [19] [30]
Meeting	Information can be gathered in formal or informal meetings and any conflict among stakeholders can handle.	Highly political/government situation, it is less effective.	Small group of stakeholders. Large size projects.	Traditional	[31] [26] [32]
Brainstorming	Different stakeholders participate in the discussion to produce many ideas and every member can show ideas about the product, whatever your idea is it can’t be criticized.	All the ideas are not qualified It might be time-consuming in case of not being organized properly. Not able to solve big issues.	Small group of stakeholders. Large/medium/small size projects.	Collaborative	[27] [21] [33] [26] [31]
Joint Application Development (JAD)/ Rapid Application Development (RAD) session	Information is gathered and decisions are made more rapidly because it is done in the presence of key stakeholders, customers, and management who share their opinions on what should be done or changed in the project.	All the participants need to be experienced and fully trained. Joint Application Development (JAD) session is expensive It requires proper planning otherwise all the effort will be of no use.	Small group of stakeholders. Small and medium size projects.	Collaborative	[21] [26] [27] [34] [25] [22]
Prototyping	It is the initial or the dummy version of the system because user can experience the strengths and weaknesses of the system.	Cost can increase than expected. Could be time-consuming for difficult and complex systems. A lot of resources are required.	Medium group of stakeholders. Large size projects.	Collaborative	[34] [26] [27] [21] [28]
Requirement Workshop	It consists of small meetings and limited participants came together for discussion, validation, and refining of under developing system’s requirements; complex requirements are gathered.	Proper planning is required Difficult to bring the stakeholders in the workshop at the same because of a busy schedule A lot of time is required for preparation	Small group of stakeholders. Large size projects. Large requirements needed to be gathered.	Collaborative	[30] [25] [34] [27] [21] [18]
Group Work	Promoting stakeholders’ commitment and cooperation because of different stakeholders. There is a moderator who keeps the participants encouraged for interaction among participants.	Expert and experienced requirement engineers are required. The schedule needed to be maintained properly and group work session is difficult to handle.	Small group of stakeholders. Large/medium/small size projects.	Collaborative	[35] [34] [27]
User Scenario	Scenarios are the stories that explain how the system can be used. It contains detail about all series of actions taken by the user.	Drawing user scenarios can be difficult. For all types of projects, User scenarios are not applicable.	Medium group of stakeholders. Small size projects.	Collaborative	[27]

Laddering	Interviews are taken to gather information like values, attributes, and goals from stakeholders by asking limited questions about the system and as well as wants, needs, and demands are prioritized. Requirements are managed in hierarchical form.	When there is a large number of requirements this technique becomes difficult. Managing and adding/deleting user requirements is complex in the hierarchy. It does not work well when a new system has to be built.	Small group of stakeholders. Medium and small size projects.	Cognitive	[27] [30] [18] [25] [36]
Card Sorting	In this technique, cards are distributed to the clients in accordance with the domain name. It is a cheap as well as a fast technique to elicit requirements.	It is not applicable for large and complex systems. Requires a limited number of cards for better performance.	Small group of stakeholders. Medium and small size projects.	Cognitive	[33] [27] [31] [36]
Repertory Grids	Phycologists use this technique and expert stakeholders are asked to evolve attributes and allocate values to the collection of domain entities.	It requires plenty of effort when information is elicited from experts. It is a time-consuming technique.	Small group of stakeholders. Large/medium/small size projects.	Cognitive	[30] [31] [33] [27]
Class Responsibility Collaboration (CRC)	Software requirements are represented by using Class Responsibility Collaboration (CRC) It can also replace UML diagrams. Users and experts work with users jointly.	Provide finite details about requirements. It is not applicable for large systems. It is time-consuming.	Small group of stakeholders. Small size projects.	Cognitive	[27] [36]
Protocol Analysis	Meeting between stakeholders and analysts will held in which user is requested to speak loudly their thought and ideas about the system.	It is time taking technique and not reliable.	Medium group of stakeholders. Small size projects.	Cognitive	[27] [31] [18] [25] [36]
Task Analysis	Different observational and interviews strategies are used to gather the description of the system which is used by experts to solve complex tasks.	Performing this task, requires a lot of concentration and effort.	Small group of stakeholders. Large size projects.	Cognitive	[31] [26]
Ethnography	Requirement engineer interacts with stakeholders and tries to become his/her friend for observing him/her to gain domain knowledge of the project.	No planned restriction. Ordinary requirement engineers can face difficulty so physiologists are required.	Small group of stakeholders. Large size projects.	Contextual	[34] [33] [37] [26]
Discourse analysis	It is the expanded version of ethnography techniques. It consists of two sub techniques known as Speech Act Analysis & Conversation Analysis.	It does not provide absolute answers to a specific question.	Medium group of stakeholders. Small size projects.	Contextual	[31]
Through Away paper prototyping	User draw features by using pen/pencil and share with requirement engineers. This paperwork can't be used for a long time.	It is suitable for medium and small-sized projects.	Medium group of stakeholders. Medium and small projects.	Innovative	[31]

B. Tools/Techniques in Agile Models (RQ2):

After discussing the Tool/Technique of elicitation phase in details. Tools/Techniques of elicitation in agile model is very important for project development. Now discussing some approaches or tools/techniques of agile models in detail. Different Tools/Techniques are in Table 2.

Table 2: Tools and techniques for Elicitation in Agile

Tool	Working	Advantages	Limitation	Ref.
User Stories	The stories explain how the system can be used in other words interaction of the user with the system. It contains detail about all series of actions taken by the user.	Short time is required for creating user stories. One copy is prevented by using user stories.	There is no aid for auto-numbering as well as deletion is difficult. Sharing is blocked when there is a single copy.	[4]
FlexREQ Model	It is used to manage requirements by using DLL (Dynamic Link list) and ER model (Entity Relation Model).	Requirements can be managed in a perfect way. Fewer chances of risk Conflicts can be Handled between requirements.	It is expensive. Large data can't be stored.	[4]
DiCOT Model	DiCOT (Distributed Cognition for Teamwork) is used to make a design system that has not been implemented yet.	It stores large amount of data. All the records can be handled effortlessly.	Data can be stolen. Trained faculty/staff is required. Backup is required.	[4]

Joint Application Development (JAD)	Information is gathered and decisions are made more rapidly in the presence of key stakeholders who share their opinions on what should be done or changed in the system.	Scrum Manager is not needed. Good Quality.	Experienced and trained staff is required. It requires lots of effort in planning and schedule.	[4]

III. Discussion

This paper deals with tools/techniques for the elicitation phase of Software Requirement Engineering (SRE) there are two types of research questions. The first research question describes overall tool/techniques in elicitation phase of Software Requirement Engineering (SRE), to find suitable and informative tools/techniques for elicitation phase of Software Requirement Engineering (SRE) approximately 38 articles were searched, there more than enough tools/techniques, the more useful and helpful tools/techniques were selected for instance interview, prototype, meeting, survey, laddering and many more. To gather information from stakeholders' interviews is taken place between two or more stakeholders. The technique in which information is gathered by asking questions is called a Questionnaire. Meetings are held to elicit information it is also convenient when there is any conflict among stakeholders. Mostly meeting, interviews, prototyping is used because these techniques are helpful in eliciting information because interaction of engineers and customer is high. In the previous discussion general tools/techniques have been described with their limitation and working which were used in any type of software development method, agile development model is different than others so tools/techniques that are used in this model have been discussed in the second research question such as user stories, Joint Application Development (JAD). The technique which defines the interaction of the user with the system is the user story it contains details about all actions taken by the user, to find or gather information rapidly and in a short time then there is no other technique better than Joint Application Development (JAD). The agile model only focuses on the project's simplicity and scope despite all this it should also work on project's problems and discovery to fill this limitation some authors agree on adding requirement elicitation phase in agile development like creative thinking [38]. This paper is very convenient for those software engineers who are looking for the most suitable and frequent tools/techniques for eliciting information in Software Requirement Engineering (SRE) as well as for the agile development model.

IV. Conclusion

The main target of this analysis is to identify the best tool/technique for the appropriate situation. For this motive, we have reviewed different available research articles of the time period from 2010-2021. We have identified 38 primary studies based on the specified key terms and keywords. For client and engineer interaction we use prefer prototype. In case of rapid requirement gathering, state-of the art prefer Joint application development. Our study thoroughly presents the strengths, weaknesses of identified approaches and model for traditional elicitation of requirements along with requirement gathering in agile development. Moreover, our approach also discuss the applicability of these tools and techniques with respect to the context of the project environment.

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